

Ames Housing Exploratory Data Analysis

End-to-End Data Analysis Capstone Report

1. Project Overview

Primary question: What property characteristics are most strongly associated with residential sale prices in Ames, and what unusual observations or data-quality issues should be considered when interpreting those relationships?

The analysis follows the project pipeline specified in the notebook: raw-data assessment, missing-data analysis, cleaning, univariate and bivariate EDA, correlation analysis, outlier investigation, multivariate analysis, hypothesis testing, findings, and limitations. The work uses 1,460 property records and 11 variables, with **SalePrice** as the target, **Neighborhood** as a categorical predictor, several numerical/ordinal predictors, and **Id** as an identifier.

2. Data Quality and Cleaning

Initial checks found **248 missing LotFrontage values** and no missing SalePrice values. There were **0 duplicate rows** and **0 duplicate Id values**. Domain checks found no invalid values for OverallQual, GrLivArea, YearBuilt, Bedrooms, FullBath, GarageCars, LotArea, SalePrice, or observed LotFrontage under the notebook's stated rules.

Metric	Result
Raw dataset	1,460 rows × 11 columns
Missing LotFrontage	248 (17.0%)
SalePrice mean / median	\$237,645 / \$230,600
OverallQual ↔ SalePrice	Pearson $r = 0.799$
GrLivArea ↔ SalePrice	Pearson $r = 0.601$; Spearman $\rho = 0.611$
OverallQual hypothesis test	Spearman $\rho = 0.7919$
Neighborhood hypothesis test	$H = 1142.9893$; $\epsilon^2 = 0.7813$

The notebook compares three LotFrontage strategies: dropping missing rows, overall-median imputation, and neighborhood-specific median imputation. Dropping retains 1,212 observed values, while median-based approaches retain all 1,460 rows. The observed LotFrontage mean is 70.02 and standard deviation is 21.90; neighborhood-median imputation produces a mean of 69.90 and standard deviation of 19.98. The final analysis therefore uses **neighborhood-median imputation**, with an overall-median fallback if required. This preserves observations while using neighborhood context.

3. Exploratory Findings

SalePrice has a mean of **\$237,645**, median of **\$230,600**, standard deviation of **\$65,093**, and ranges from **\$92,100 to \$595,000**. Its skewness is **0.667**, indicating a right-tailed distribution rather than a perfectly symmetric one.

Among the numerical variables examined, **OverallQual** has the strongest correlation with SalePrice ($r = 0.799$), followed by **GrLivArea** ($r = 0.601$), FullBath (0.396), GarageCars (0.326), YearBuilt (0.272), Bedrooms (0.254), and LotArea (0.101). LotFrontage has a near-zero correlation with SalePrice (-0.011). Correlation describes association rather than causation.

The GrLivArea/SalePrice relationship is positive and broadly monotonic. Pearson correlation is **0.601**, while Spearman correlation is **0.611**; the similarity supports a broadly consistent relationship, although the scatter contains deviations and extreme cases.

Sale prices also vary substantially across neighborhoods. Median SalePrice is highest in the observed neighborhood summary for NoRidge (\$373,000), NridgHt (\$367,700), and StoneBr (\$363,300), while lower observed medians include MeadowV (\$167,450), IDOTRR (\$169,700), and BrDale (\$175,100). These are descriptive differences in this dataset and should not be interpreted as causal neighborhood effects.

4. Quality, Size, and Outliers

Quality bands show a clear descriptive price gradient: median SalePrice is \$179,350 for Low quality (1–4), \$235,600 for Medium (5–6), \$296,900 for High (7–8), and \$371,550 for Very high quality (9–10). The multivariate analysis then examines whether the GrLivArea/price relationship changes across these quality groups and combines Neighborhood and QualityBand using a median-price heatmap.

The IQR procedure identifies 8 potential GrLivArea outliers, 11 LotArea outliers, and 6 SalePrice outliers. Two especially large living-area observations exceed 3,500 square feet: Id 1016 in Timber (4,600 sq ft, \$160,000) and Id 858 in IDOTRR (4,500 sq ft, \$145,000). The notebook correctly treats extreme values as records requiring investigation rather than automatically deleting them.

5. Hypothesis Testing

Test 1 — OverallQual and SalePrice. The notebook uses Spearman rank correlation because OverallQual is an ordinal/discrete quality measure and a monotonic relationship is of interest. The result is $\rho = 0.7919$ with a p-value below the displayed machine precision threshold. The effect size is the same rho value, indicating a strong positive monotonic association in this sample. The interpretation should consider the effect size and diagnostic plots rather than the p-value alone.

Test 2 — Neighborhood and SalePrice. The Kruskal-Wallis H test is used for the multiple independent neighborhood groups. The result is $H = 1142.9893$, $p = 3.11 \times 10^{-23}$, with epsilon-squared = 0.7813 . Under the notebook's hypotheses, the evidence is consistent with differences in sale-price distributions across neighborhoods. The result does not establish that neighborhood itself causes price differences.

6. Multivariate Interpretation and Key Takeaways

The analysis shows that SalePrice is associated with several property characteristics, with OverallQual and GrLivArea standing out among the numeric predictors. Neighborhood context is also important descriptively: the neighborhood-level distributions differ, and the combined Neighborhood $\times$ QualityBand analysis shows that price patterns depend on more than one property characteristic. The quality-band analysis further indicates that the size-price relationship should be interpreted in the context of property quality.

A useful data-quality observation is that LotFrontage missingness is concentrated unevenly across neighborhoods. Because missingness is not simply a random deletion of rows, the notebook compares imputation strategies before selecting a neighborhood-aware approach. This keeps the full sample while limiting the assumption that every neighborhood has the same frontage distribution.

The notebook also includes an explicit unexpected-finding investigation section. It identifies candidate findings such as weak/strong relationships, subgroup changes, extreme properties, or concentrated missingness and instructs the analyst to select one from the actual outputs rather than inventing a result. The notebook does not record a separate final prose conclusion naming one unexpected finding, so this report does not claim one beyond the documented outputs.

7. Limitations

First, the analysis is observational, so statistical associations should not be interpreted as causal effects. Second, LotFrontage required imputation, and the chosen method can affect the resulting distribution. Third, extreme observations can influence correlations and summary statistics even when they are legitimate properties. Fourth, neighborhood group sizes differ, which matters when comparing distributions. Finally, the findings come from this Ames housing dataset and may not generalize to other cities, time periods, or housing markets.

8. Conclusion

The completed EDA provides a reproducible view of the factors associated with Ames residential sale prices. Overall quality shows the strongest numeric association with SalePrice, living area has a substantial positive association, and sale-price distributions differ across neighborhoods. The project combines data-quality checks, transparent missing-data handling, purposeful visualizations, outlier review, multivariate analysis, and effect-size-aware hypothesis testing. These results support descriptive understanding of the dataset while maintaining the important distinction between association and causation.

Reproducibility: The analysis is implemented in the supplied Jupyter notebook Ames_Housing_EDA_Capstone.ipynb. The final validation checks confirm 1,460 raw rows, 11 columns, unique Id values, non-missing SalePrice, and zero remaining LotFrontage values after cleaning.